

JEE MAIN REVISION MASTER: CIRCULAR MOTION

2026 0 QUESTIONS 0% WEIGHTAGE ↓100%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 7 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Circular Motion - Core Application	7 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Circular Motion - Core Application (Moderate Yield — 7 PYQs)

Core Principle: Key problem pattern in Kcet Circular Motion.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2+y^2=25$, where x and y are in meter. The ..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An athlete runs along a circular track of diameter 80 m . The distance travelled and the magnitude of displacement of the athlete when he co..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The speed of the particle is 2ms^{-1} , when the particle attains the lowest 'y' co-ordinate, the acceleration of the particle is (in ms^{-2})

- (A) $0.4\hat{j}$
- (B) $0.8\hat{i}$
- (C) $0.8\hat{j}$
- (D) $0.4\hat{i}$

Correct Answer: None

Solution Analysis: To determine the acceleration of a particle in uniform circular motion, we start with the equation of the circle: $(x-2)^2 + y^2 = 25$. This equation represents a circle with its center at $(2,0)$ and a radius r of 5 meters, derived from: $r^2 = 25 \Rightarrow r = 5$. In uniform circular motion, the centripetal acceleration a_c is given by the formula: $a_c = \frac{v^2}{r}$. Substituting the given speed $v = 2\text{m/s}$ and the radius $r = 5\text{m}$: $a_c = \frac{2^2}{5} = \frac{4}{5} = 0.8\text{m/s}^2$. When the particle is at the lowest y-coordinate, it is at the point $(2, -5)$, assuming